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Introduction to Biophysics Winter Semester 2025/26, Exercise Sheet 11

Problem 11.1 Membrane transport and membrane potential

Which of the following statements are true? Explain why or why not?

- ☐ Transport by transporters can be either active or passive, whereas transport by channels is always passive.
- ☐ Transporters saturate at high concentrations of the transported molecule when all their binding sites are occupied; channels, on the other hand, do not bind the ions they transport and thus the flux of ions through a channel does not saturate.
- ☐ The membrane potential arises from movements of charge that leave ion concentrations practically unaffected, causing only a very slight discrepancy in the number of positive and negative ions on the two sides of the membrane.

Problem 11.2 Changing the membrane potential via ion transport

Consider a spherical cell with radius $R = 100 \mu\text{m}$. The capacitance of the cell membrane is $1 \mu\text{F}/\text{cm}^2$. How much charge needs to be transported across the membrane in order to change the voltage across the membrane by 100 mV? If this charge transfer occurs via monovalent ions, what is the corresponding change in intracellular ion concentration? How does this concentration change compare to typical absolute concentrations of intracellular ions? How does this change, if the cell has a radius of only $R = 1 \mu\text{m}$?

Problem 11.3 Membrane potential by Na^+ and K^+

The squid giant axon occupies a unique position in the history of our understanding of cell membrane potentials and nerve action. When an electrode is stuck into an intact giant axon, the membrane potential registers -70 mV . When the axon, suspended in a bath of seawater, is stimulated to conduct a nerve impulse, the membrane potential changes transiently from -70 mV to $+40 \text{ mV}$.

For univalent ions and at 20°C (293 K), the Nernst equation reduces to

$$V = 25.2 \text{ mV} \ln \frac{C_o}{C_i}$$

where C_o and C_i are the concentrations outside and inside, respectively.

Using this equation, calculate the potential across the resting membrane (1) assuming that it is due solely to K^+ and (2) assuming that it is due solely to Na^+ . (The Na^+ and K^+ concentrations in axon cytoplasm and in seawater are given in Table 1.) Which calculation is closer to the measured resting potential? Which calculation is closer to the measured action potential? Explain why these assumptions approximate the measured resting and action potential.

ION	CYTOPLASM	SEAWATER
Na ⁺	65 mM	430 mM
K ⁺	344 mM	9 mM

TABLE 1: Ionic composition of seawater and of cytoplasm from the squid giant axon.